QP CODE: 25900299

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC)–MG3VACCND200-HOLISTIC NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** mark

1. Option C – 60-70%
2. Option B - Phytochemicals
3. Option A – Polyphenols
4. Option A – Emotional support and belonging
5. Option B – Donating electrons to free radicals
6. Option D - Caffeine and dietary Nitrates

Part B

Short Answer Type Questions.

Answer any **2** questions.

Each question carries **2** marks.

7. **Cite how soluble fibre helps in lowering blood cholesterol.**
   - Soluble fibre helps in lowering LDL "bad" cholesterol. **(1 mark)**
   - Soluble fibre binds bile acids and cholesterol, increasing their excretion and reducing LDL cholesterol, this protects against heart disease. **(1 mark)**
8. **Discuss on Galacto oligosaccharides. (each point carries ½ marks)**
   - Synthesized from lactose or derived from lactulose.

- Strongly stimulate Bifidobacteria and Lactobacilli.
- Particularly beneficial in early infancy.
- Found in dairy products, legumes (e.g., lentils, chickpeas), and some vegetables

9. State the importance of flavonoids in cardiovascular health.

- Flavanoids block the angiotensin-converting enzyme (ACE) that raises blood pressure,by blocking the "suicide"enzyme cyclo-oxygenase that breaks down prostaglandins,they prevent platelet stickiness and hence platelet aggregation **(1 mark).**
- Flavanoids protect the vascular system and strengthen the tiny capillaries that carry oxygen and essential nutrients to all cells. **(1 mark)**

10. State one way in which prebiotics help improve cardiovascular health.

- Prebiotics may help lower the levels of blood triglycerides, a type of fat that increases the risk of heart disease**. (1 mark)**
- By improving lipid metabolism and reducing inflammation, prebiotics can support cardiovascular health and may help prevent conditions like atherosclerosis, which is the hardening of arteries due to plaque buildup. **(1 mark)**

Part C

Short Essay Type Questions.
Answer any **4** questions.
Each question carries **5** marks.

11. Explain how does time management help in reducing stress?

Effective time management is a key tool for **stress prevention** and maintaining a **holistic lifestyle**.

Techniques: (3 marks)

- **Prioritization**: Use tools like the **Eisenhower Matrix** (urgent vs. important) to focus on tasks that truly matter.
- **Goal Setting**: Set SMART (Specific, Measurable, Achievable, Relevant, Time-bound) goals for better direction.
- **Planning**: Use **daily planners, calendars, or apps** to organize tasks and deadlines.
- **Time Blocking**: Assign specific time slots for tasks to avoid multitasking and distractions.

- **Avoid Procrastination**: Break large tasks into smaller, manageable steps to overcome inertia.
- **Set Realistic Limits**: Learn to say "no" when overcommitted.

Benefits: (2 marks)

- **Reduces stress and panic** by avoiding last-minute rushes.
- Increases **productivity and work-life balance**.
- Provides a **sense of control and achievement**.
- Frees up time for **self-care and relaxation activities**.

12. Classify carotenoids and mention any two health benefits.

Classification- **3 marks**

Health benefits **- 2 marks**

1. CAROTENOIDS	
Components	**Potential Health Benefit**
Beta Carotene	• neutralizes free radicals which may damage cells • bolsters cellular antioxidant defences. • can be made into Vitamin A in the body
Lutein, Zeaxanthin	• supports maintenance of eye health
Lycopene	• supports maintenance of prostate health.

13. Explain hoe nutraceuticals help in prevention and management of obesity.

Role of Nutraceuticals in the prevention and management of Obesity

- **Appetite Suppression and Satiety Enhancement:** High viscosity fibres such as psyllium and glucomannan act as bulking agents, slowing gastric emptying and suppressing hunger. **(1 mark)**
- **Metabolic Rate Enhancement and Fat Oxidation:** Nutraceuticals like green tea extract (Epigallocatechin Gallate, EGCG), curcumin and L-carnitine increase fat oxidation and improve mitochondrial function, thereby boosting metabolism **(1 mark)**
- **Fat Absorption Reduction:** Chitosan, derived from shell fish, can bind to dietary fats in the digestive tract, reducing fat absorption and aiding in weight loss. **(1/2 mark).** Linoleic acid and capsaicin stimulate lipolytic enzymes and enhance fat breakdown **(1/2 mark).**

- **Reducing Inflammation and inducing insulin sensitivity:** Omega-3 fatty acids inhibit adipocyte markers, reduce visceral fat and improve glucose metabolism **(1/4 mark).**

  Berberine shows lipid control and insulin sensitivity **(1/4 mark)**.

  Alpha lipoic acid improves mitochondrial function and reduces adipogenesis **(1/4 mark)**

  Curcumin possess anti -inflammatory properties and can modulate hormones related to fat storage and metabolism **(1/4 mark).**
- **Modulating Gut Microbiota and promoting satiety**

  Probiotics and Prebiotics can alter the composition of gut bacteria, which plays a role in energy extraction and fat storage. **(1/2 mark).**

  Intestinal bacteria play a vital role in energy balance and appetite regulation. **(1/2 mark).**

14. Discuss on minerals with nutraceutical importance.

- **Folic acid-**Produce the genetic material of cells,in pregnancy for preventing birth defects,RBC formation,protects against heart disease.
- **Calcium**-Vital for bone and teeth health and maintaining bone strength.Important in nerve and muscle functions
- **Iron**-Vital for energy production,carry and ttransfer oxygen to tissues
- **Magnesium**-Healthy nerve and muscle function and bone formation,may help prevent premenstrual syndrome(PMS)
- **Phosphorus** -Strong bones and teeth,helps in formation of genetic material,energy production and storage.
- **Chromium**-With insulin helps to convert carbohydrates and fats into energy
- **Cobalt**- Essential component of Vitamin B12
- **Copper**-Essential for haemoglobin and collagen production,healthy functioning of the heart,energy production,absorption of iron from digestive tract.
- **Iodine**-Essential for proper functioning of the thyroid
- **Vitamin A**-Antioxidant,essential for growth and development and in the treatment of certain skin disorders. **(1/2 mark for each point)**

15. Explain the functions and sources of carotenoids and lycopene.

i. **CAROTENOIDS (2.5 marks)**

Carotenoids are pigments responsible for the red, yellow, and orange color in many fruits and vegetables. They are powerful antioxidants and some, like beta-carotene, are converted into vitamin A in the body.

- **Functions:** Eye health, immune support, antioxidant activity.
- **Sources:** Carrots, sweet potatoes, mangoes, spinach, pumpkin.

ii. LYCOPENE (2.5 marks)

Lycopene is a red-coloured carotenoid with strong antioxidant properties. It has been studied for its protective role against prostate cancer and heart disease.

- **Functions:** Protects against cancer and cardiovascular diseases.
- **Sources:** Tomatoes, watermelon, pink grapefruit.

16. Classify antioxidants with examples.

Classification **– 4 marks,** Examples **– 1 mark**

Antioxidants can be classified into two major types based on their origin:

A. ENDOGENOUS ANTIOXIDANTS (Produced within the body)

These are naturally synthesized by the body to protect against oxidative damage. They include:

- **Enzymatic antioxidants**:
    - **Superoxide dismutase (SOD)**: Converts harmful superoxide radicals into less harmful molecules.
    - **Catalase**: Breaks down hydrogen peroxide into water and oxygen.
    - **Glutathione peroxidase**: Neutralizes peroxides and protects membranes from lipid peroxidation.
- **Non-enzymatic antioxidants**:
    - **Glutathione**: A tripeptide present in most cells; acts as a powerful detoxifier and antioxidant.
    - **Uric acid** and **bilirubin** also act as natural antioxidants in the body.

B. EXOGENOUS ANTIOXIDANTS (Obtained from diet)

These are **antioxidants we get from food**, especially fruits, vegetables, nuts, and whole grains. Examples include:

- **Vitamin C (Ascorbic acid)** – Found in citrus fruits, amla, guava, bell peppers.
- **Vitamin E (Tocopherol)** – Found in nuts, seeds, vegetable oils, and green leafy vegetables.

- **Carotenoids** – Like **beta-carotene**, **lycopene**, **lutein** found in carrots, tomatoes, and spinach.
- **Polyphenols and flavonoids** – Found in tea, cocoa, grapes, berries, apples.
- **Selenium and zinc** – Trace minerals that work as cofactors in antioxidant enzyme systems.

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the functions and sources of curcumin, tannins and resveratrol.

I. CURCUMIN (3.5 marks)

Curcumin is the active compound in turmeric. It is widely known for its anti-inflammatory, antioxidant, and healing properties.

- **Functions:** Reduces inflammation, boosts brain function, supports joint health.
- **Sources:** Turmeric root.

II. TANNINS (3.5 marks)

Tannins are astringent compounds that have antimicrobial and antioxidant properties. However, in high amounts, they can interfere with nutrient absorption.

- **Functions:** Prevent bacterial growth, reduce inflammation, but may bind iron.
- **Sources:** Tea, coffee, grapes, legumes.

III. RESVERATROL (3 marks)

A compound found in the skin of red grapes and berries; resveratrol is associated with heart health and anti-aging effects.

- **Functions:** Antioxidant, improves blood flow, anti-inflammatory.
- **Sources:** Red grapes, wine, blueberries, peanuts.

18. Record the significance of Polyphenols, Spices, Functional Fungi and Omega-3 fatty acids as immune supportive nutraceuticals

Polyphenols, Spices and Functional Fungi

- **EGCG (green tea)**-shows antiviral activity, enhances T-cell function and reduces influenza risk by 33% **(2 marks)**
- **Allicin(garlic)**-Antiviral and anti-inflammatory, supporting innate defense via natural killer and macrophage activation **(2 marks)**

- **Curcumin, quercetin, resveratrol**-Exhibit antioxidant, anti-viral and immune modulating effects **(2 marks)**
- **Functional Mushrooms** (eg: Active Hexose correlated compound, reishi, turkey tail)-Activate TLR (Toll-like receptor) pathways, enhance NK and T-cell activity and reduce infection severity in animal studies **(2 marks)**

Omega-3 Fatty acids

- **EPA/DHA** promote anti-inflammatory resolvins, reduce pro-inflammatory cytokines and support innate immunity by maintaining cell membrane integrity. **(2 marks)**

19. Describe in detail the internal (endogenous) and external (exogenous) processes processes that contribute to free radical production.

A. Internal (Endogenous) Sources of Free Radicals (5 marks)

These originate within the body during normal physiological processes.

- **Normal Metabolic Processes:**
  - Occur during cellular respiration in mitochondria.
  - Partial reduction of oxygen produces Reactive Oxygen Species (ROS) such as *superoxide radical ($O_2\bullet^-$)*.
- **Immune System Response:**
  - White blood cells (e.g., macrophages) produce free radicals like *nitric oxide* and *superoxide* to destroy pathogens.
  - Excess production can damage body tissues.
- **Inflammation:**
  - Inflammatory reactions release enzymes and reactive compounds that generate free radicals.

B. External (Exogenous) Sources of Free Radicals (5 marks)

These are generated from external environmental or lifestyle factors.

- **Pollution:**
  - Air pollutants (vehicle exhaust, industrial smoke) introduce chemicals that trigger free radical formation.
- **Cigarette Smoke:**
  - Contains thousands of chemicals and free radicals.
  - Promotes free radical production in lungs and bloodstream.
- **Radiation (UV rays, X-rays):**

    - Splits molecules in skin and tissues, forming radicals.
    - Leads to skin aging, DNA damage, and cancer risk.
- **Processed Foods and Alcohol:**
    - High-fat, high-sugar foods and alcohol metabolism generate free radicals and increase oxidative stress.
- **Certain Medications and Chemicals:**
    - Drugs (e.g., chemotherapy), pesticides, and toxins increase radical production during metabolism or tissue interaction.

20. Discuss the definition, goals and importance of holistic nutrition.

Definition:

"Holistic nutrition is a comprehensive approach to nutrition that considers the individual's entire well-being, including physical, mental, emotional, and spiritual aspects, rather than just focusing on dietary restrictions or specific nutrient needs. It emphasizes a personalized approach to health, taking into account individual needs, preferences, and lifestyle factors." **(3 marks)**

Goals: (3.5 marks)

- To support complete health—physical, mental, emotional, and spiritual.
- To prevent diseases and promote long-term well-being.
- To nourish the body using whole, natural, and unprocessed foods.
- To provide personalized nutrition based on individual needs and preferences.
- To encourage a healthy and mindful relationship with food.
- To create harmony between food, lifestyle, and the environment.

Importance: (3.5 marks)

- It improves total wellness by focusing on the whole person, not just physical health.
- It promotes better lifestyle choices including sleep, exercise, and stress control.
- It helps reduce stress by supporting emotional balance and mental clarity.
- It reduces the use of unhealthy, processed foods by encouraging natural eating.
- It boosts digestion and strengthens the immune system.
- It supports sustainable living and responsible use of natural resources.